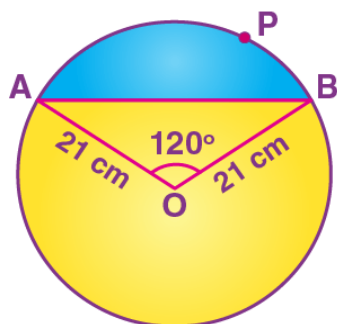


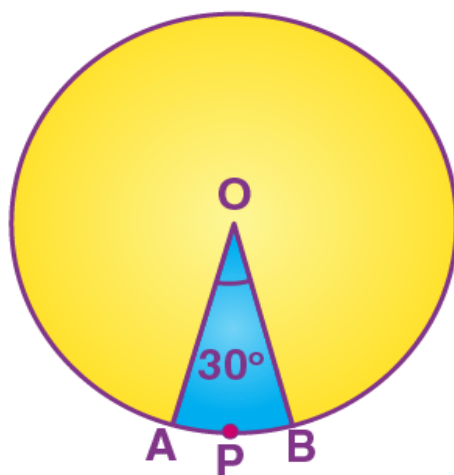
Class 10 MATHEMATICS (Circle & Area related to circle)

Q.1: Find the area of the segment APB shown in the figure, if the radius of the circle is 21 cm and $\angle AOB = 120^\circ$. (Use $\pi = 22/7$).



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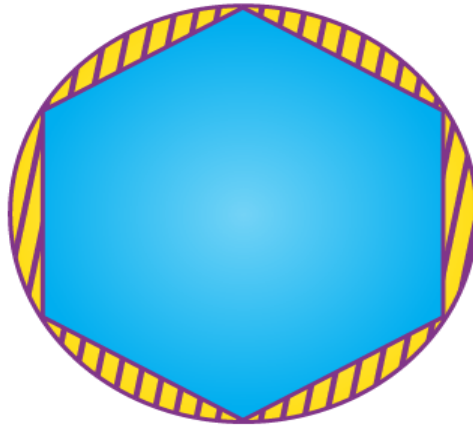
Q.2: Find the area of the sector of a circle with a radius of 4 cm and of angle 30° . Also, find the area of the corresponding major sector (Use $\pi = 3.14$)



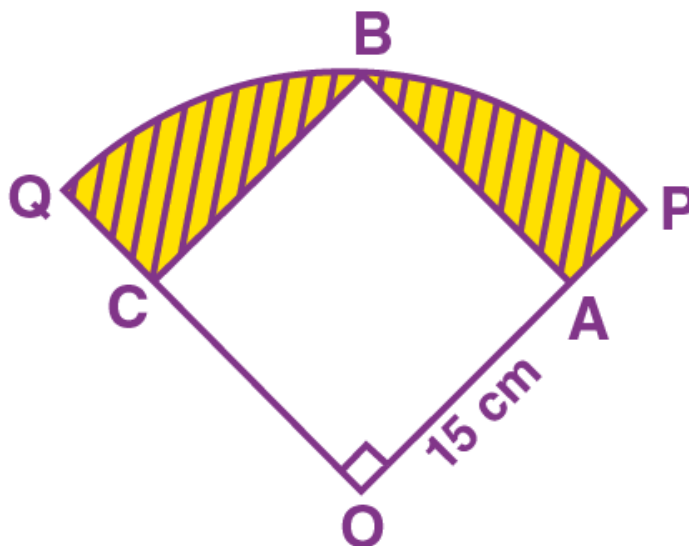
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Q. 3 The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

Q.4: A round table cover has six equal designs as shown in the figure. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of Rs. 0.35 per cm^2 . (Use $\text{root } 3 = 1.7$)



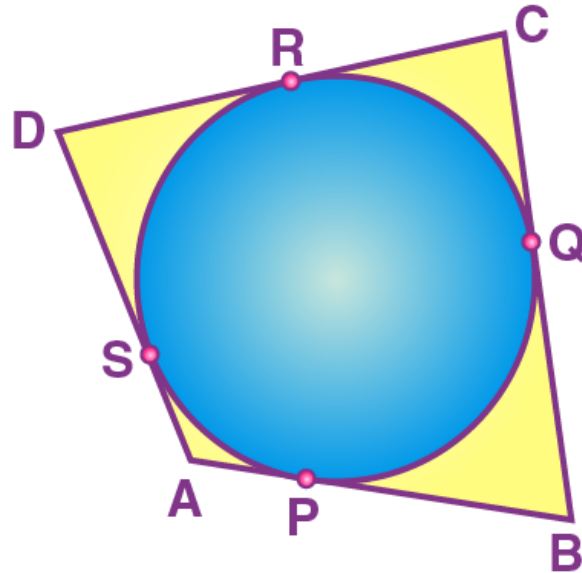
Q. 5 In the figure, a square OABC is inscribed in a quadrant OPBQ. If OA = 15 cm, find the area of the shaded region. (use $\pi = 3.14$)



Q. 6. If the area of a circle is numerically equal to twice its circumference, then the diameter of the circle is A) 4 units B) π units C) 8 units D) 2 units

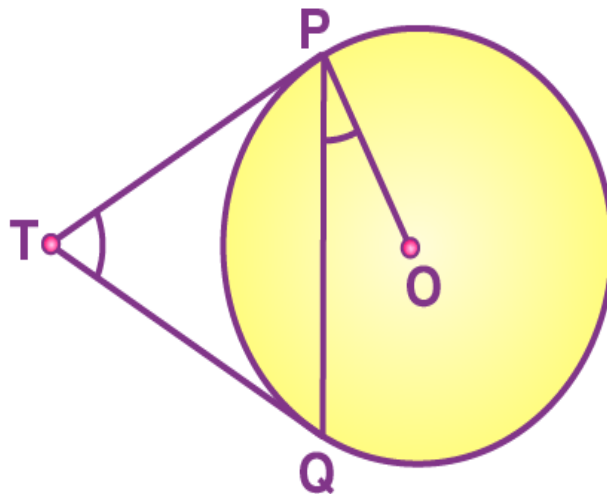
Q. 7: Prove that the tangents drawn at the ends of a diameter of a circle are parallel.

Q. 8: A quadrilateral ABCD is drawn to circumscribe a circle as shown in the figure. Prove that $AB + CD = AD + BC$



Q. 9: Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

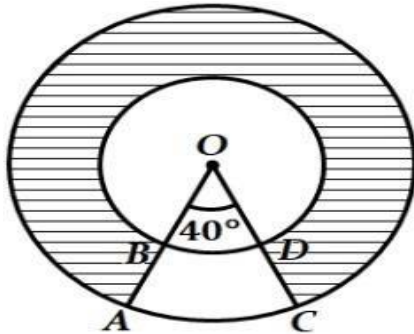
Q.10: In the figure, two tangents TP and TQ are drawn to a circle with centre O from an external point T, prove that $\angle PTQ = 2\angle OPQ$.



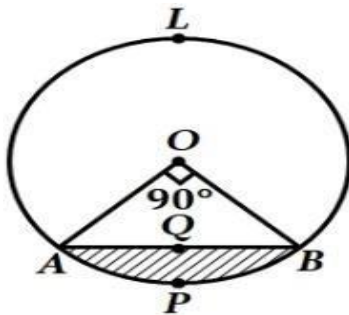
Q.11: Prove that the lengths of tangents drawn from an external point to a circle are equal.

Q.12: Prove that the tangent drawn at the mid-point of an arc of a circle is parallel to the chord joining the endpoints of the arc.

Q. 13 In figure , find the area of the shaded region, enclosed between two concentric circles of radii 7 and 14



Q14 . In figure, is a chord of circle, with centre O and radius 10cm , that subtends a right angle at the centre of the circle. Find the area of the minor segment AQB . Hence find the area of major segment ALBQA. (Use $\pi = 3.14$)



Q.15 Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact.

